GLOBAL STUDY SUPPORT / REFERENCE SERIES

Conceptual Bracket Assembly

ILLUSTRATIVE DEMONSTRATION - NOT CLIENT WORK

Objective

Introduce the questions involved in a simple bracket-and-plate assembly. This is a workflow reference, not a finished CAD model or a validated mechanical design.

Illustrative assumptions

Consider a flat base plate of 100 x 60 x 8 mm and an upright plate of 60 x 50 x 6 mm. Material is unspecified. Dimensions are invented to support discussion and must not be treated as a specification.

Modelling sequence

Sketch the base profile and extrude. Create a second part for the upright. Add mounting features only after defining the intended load path and edge distances. Assemble using fully defined, non-redundant mates. Review interference and access for fasteners.

Design review checklist

Confirm units, intended loads, support conditions, material properties, tolerances, manufacturing constraints and fastening method. A simulation needs a justified mesh and boundary conditions before its results are meaningful.

Available deliverable and limits

This demonstration contains reference notes and a PDF only. No native SolidWorks assembly, manufacturing drawing or simulation result is supplied. Seek competent engineering review before any physical use.